01 / MISSION

Haze Agent Harness Architecture

Version 0.8 | 22 October 2026 | Six-day delivery and two paths to reducing haze harm

Mission

Investigate haze harm, explain evidence and causes, compare feasible interventions, and test a clearly defined 10% reduction target. Build a reusable evidence-to-action workflow while validating each scientific model and intervention for its own scope.

One executable pilot; two expansion paths

Layer	v0.8 design boundary
Hackathon	One outdoor maintenance crew in Klang, Selangor; plan_outdoor_work for 08:00-12:00 local. Compare feasible schedules using a named modeled PM2.5 exposure proxy.
Worker protection	Extend to an employer-led field pilot with actual activity, implementation and suitable exposure measurements.
Source prevention	Discover a landscape/operator/funder partnership; connect preventive actions to verified delivery and evaluated consequences.
Shared foundation	Four departments reuse Manager, HazeOps, WorkerShield and EvidenceCritic; Zetaris evidence reads; local H-MEM and World Model; LangGraph execution.
Broader damage	Add scoped health, direct-loss, disruption and climate accounting as reviewed methods and inputs become available. Keep other harms and missing coverage visible.

What this revision adds

A six-day delivery contract; equal-resource planner comparisons; a real workplace pilot protocol; source-prevention planning and delivery contracts; a scoped damage ledger; causal evaluation and activation gates.

Status: design specified, not demonstrated. Source access, implementation, partnerships, scientific review and acceptance execution remain pending. A 10% reduction is an objective to test; modeled crew exposure does not establish regional total-damage reduction.

02 / READING MAP

Read by delivery stage

The hackathon scope and the wider impact programme have explicit, separate evidence requirements.

Area	Pages	Purpose
Mission and organization	3-10	System map, target contract, departments and continuing coordination.
Evidence and models	11-14	Source monitoring, canonical evidence, exposure/impact tools and scenarios.
User and pilot workflows	15-19	Workspace, Klang crew workflow and later specialist capabilities.
Execution and integrations	20-26	LangGraph, bounded peers, sponsors, packaging, interfaces and audit.
Six-day delivery	27-29	One connected demonstration, acceptance evidence and daily exit results.
Expansion programme	30-36	Two impact paths; workplace evaluation; prevention delivery; damage ledger; causal comparison and activation.
Sources and revision	37-41	Retained references, expansion evidence, integrated changes and readiness.

Three scopes, one evidence foundation

Hackathon: the existing four-role Klang workflow, deterministic baseline/scenario exposure comparison, one meaningful evidence update and inspectable explanation. Use pp.16 and 27-29 as the delivery boundary.

Next operational tests: an employer-led workplace pilot and, in parallel, source-prevention partner discovery. Each needs an accountable operator, feasible actions, resources and its own comparison strategy.

Regional ambition: evaluated prevention plus reviewed damage accounting. Expanded geography, a new profile or a faster update cadence does not establish effectiveness or complete damage coverage.

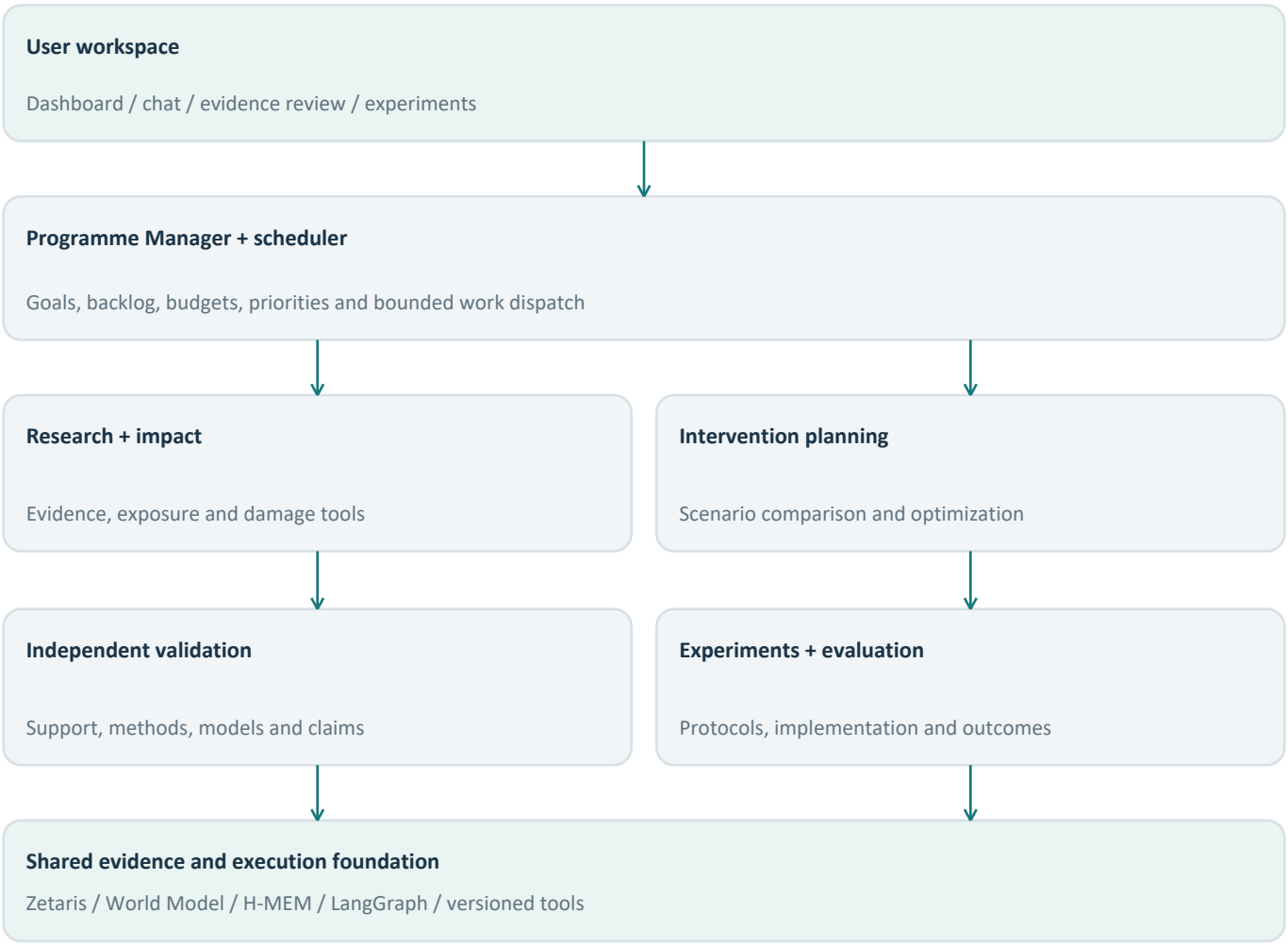
> **Status vocabulary**
>
> **Specified** means described. **Enabled** means configured. **Executed** means evidenced by a captured run. **Validated** means reviewed for a stated use. These states are not interchangeable.

v0.8 integrates the approved expansion review into the architecture. It retains v0.7 engine ownership, independent review, exact evidence revisions, offline path and sponsor-integrity requirements. The original v0.7 source is preserved.

03 / SYSTEM MAP

Departments over a shared harness

A department owns a continuing responsibility. A specialist run performs a bounded task.



Department arrows represent coordination, not four mandatory model instances. Existing HazeOps and WorkerShield capabilities, deterministic tools and review executions can serve several departments. Manager and EvidenceCritic remain crosscutting roles.

Source watchers and the programme scheduler create bounded jobs. LangGraph owns the transitions, checkpoints and recovery inside each job. The programme queue is a work backlog, not a competing graph-node retry system.

Two different model meanings

World Model owns canonical evidence and decision state. Physical, exposure, health, economic and intervention models are separate versioned scientific tools. Storing a relationship does not validate its scientific correctness.

v0.8 expansion paths

Worker protection and source prevention share this foundation. Their operational protocols and evidence gates are specified on pp.30-36; they do not add four mandatory model instances.

04 / SCOPE AND TARGET

Define the 10% contract

A percentage needs a comparable baseline, metric, population, period and accounting boundary.

Outcome	Meaning	Supported claim
Ambient pollution	Outdoor concentration or a defined fire-attributable environmental measure.	A change in the named measure, with source attribution where claimed.
Human exposure	Concentration encountered by a defined population across locations and activities.	A modeled or measured exposure change; health benefit needs its own model.
Damage	Specified health, economic, environmental or climate consequences.	A scoped estimate/evaluated change using an applicable accounting method.

Hackathon design default

Use one Klang maintenance crew, 08:00-12:00 local, and the concentration-time exposure proxy on p.13. Compare feasible plans for the same required work and environmental scenario. Test whether at least 10% projected improvement is supported; infeasibility is a valid result.

Record before enabling target claims

target_id/version; owner and objective; metric/units; geography and population; fire/event and assessment periods; baseline; accounting coverage; available actions; budget; constraints; method/model versions; uncertainty; review and measurement plan. Confirm the complete programme contract before operational use.

Use a separate contract for every expansion

A workplace field pilot retains its specified crew metric. A prevention programme defines treated sources/landscape and affected populations, including downwind effects. A damage programme selects categories and valuation rules. Broader ambitions do not change the first pilot's claim boundary.

Comparable boundaries and claims

Include rescheduled work, overtime, lost service, costs and transferred exposure across the full assessment horizon. Separate target, projected reduction, implementation, measured change and attributable effect. A zero, invalid or incomparable baseline cannot support a percentage.

05 / ORGANIZATION

Four continuing departments

Organize work by scientific responsibility while reusing the existing specialist profiles.

Department	Owns	First-pilot support
Research and Impact Measurement	Questions, source discovery, evidence applicability, exposure estimates and scoped damage models.	HazeOps, Manager-routed research tasks and versioned calculation tools.
Intervention Discovery and Planning	Candidate actions, comparable scenarios, feasibility and projected progress toward the selected target.	WorkerShield with explicit scheduling, scenario and optimization tools.
Independent Validation	Independent checks of source support, methods, assumptions, models and conclusions.	EvidenceCritic, deterministic checks and appropriate domain review.
Experiments and Evaluation	Hypotheses, test protocols, implementation records, observations and comparative outcome reports.	Manager-coordinated tests with HazeOps/WorkerShield artifacts and an experiment registry.

Department charter

Each charter declares its goal, supported scope, backlog, source and tool access, output artifacts, review requirements, budgets, cadence and stopping conditions. It persists between individual worker runs.

Coordination and independence

The Programme Manager assigns work and resolves dependencies. Researchers and planners can request validation, but cannot approve their own scientific changes. Experiment owners produce records; independent validation reviews design and interpretation.

The four departments are new programme responsibilities, not a claim that four additional agents exist. HazeInfo and the other later domain profiles remain later until enabled and evaluated; their absence does not prevent scoped research tools from supporting the pilot.

Core loop

Research -> candidate interventions -> reviewed experiments -> measured outcomes -> updated research. Independent validation applies throughout, and every handoff retains exact artifact and evidence revisions.

06 / RESEARCH DEPARTMENT

Research and impact measurement

Maintain the best supported estimate for a stated question, place, population and period.

Continuing responsibilities

Maintain research questions, identify missing evidence, monitor relevant sources, compare findings and produce reproducible exposure or impact estimates. Keep competing explanations and unresolved uncertainty visible.

Building block	Required behavior
Source discovery	Search approved research APIs, official sources, environmental feeds and relevant news. Full-text access, licences, rate limits and source coverage are explicit.
Paper extraction	Preserve title, DOI/URL, version, publication and retrieval times, exact passage, section/page when available, study design, population, geography, conditions and limitations.
Applicability review	Determine whether a finding applies to the pilot. Distinguish a reported association, causal claim, model assumption and intervention-effectiveness estimate.
Impact tools	Use reviewed numerical methods for exposure and selected damage categories. Return input revisions, method version, outputs, uncertainty and missing inputs.

Research evidence lifecycle

Discovered -> extracted/normalized -> candidate -> reviewed -> accepted for a stated use. Material can instead be rejected, superseded, withdrawn or stale. Acceptance has a scope and reviewer; it is not universal scientific endorsement.

News can identify events or research leads. Retrieve the underlying study or official record when a quantitative claim needs stronger support. Preserve conflicting findings rather than merging them into apparent consensus.

Deliverable

A research finding or impact estimate links the question, exact sources, calculation method, applicability, supporting/conflicting evidence, limitations, review status and the decisions it may affect.

Example: a new study suggests different filtration effectiveness. Extract it as candidate evidence, review its setting, then request a model update only if applicable. Publication alone does not change an accepted parameter.

07 / INTERVENTION DEPARTMENT

Discover and compare interventions

Find feasible actions and estimate their benefit against the same recorded baseline.

Continuing responsibilities

Maintain an intervention catalogue, compare evidence of effectiveness, identify practical constraints and generate candidate plans. Track which assumptions determine whether the selected target can be reached.

Service	Output
Intervention catalogue	Action, mechanism, eligible population, supporting research/guidance, required resources, applicability limits and known implementation dependencies.
Scenario simulator	Baseline and action estimates using compatible evidence, model versions and assessment boundaries. Show resources, expected effect and uncertainty.
Constrained optimizer	Rank feasible combinations under budget, staffing, task, timing, capacity and access constraints. Include distribution of benefit and unmet needs.
Recommendation assembly	WorkerShield turns the supported comparison into a reviewable crew proposal. Later HazeShield can apply the same services to community allocations.

What the first pilot compares

Compare the original crew schedule with available timing, task-order or control alternatives. Record the actual workplace facts required by a scenario. Do not invent exact safe durations, available equipment or an approved closure.

Report environmental changes, exposure changes and damage changes under their own metric names. Include implementation costs and operational disruption. Do not simply add percentage benefits from overlapping controls.

Outcomes the planner can return

Supported candidate; candidate requiring more evidence; target infeasible under current constraints; or insufficient evidence for a quantified recommendation. Explain which constraint or uncertainty changes the ranking.

Deliverable

A versioned scenario set and proposed plan, with baseline, actions, feasibility, projected reduction, sensitivity, assumptions, review requirements and the next useful test.

Expanded intervention contract

Later prevention actions also record operator/authority, funder, full cost, delivery probability, verified completion and additional avoided harm. Use pp.32-33 and account for overlapping actions.

08 / VALIDATION DEPARTMENT

Independent scientific validation

Expand evidence critique while keeping runtime permission and scientific support separate.

Review layer	What it checks
Source and claim support	Exact evidence revisions, passage-to-claim support, source quality, conflicting findings, corrections, scope and freshness.
Method and model applicability	Population, geography, exposure period, study design, model assumptions and whether the method supports the claimed inference.
Calculation correctness	Reproducible inputs, units, aggregation, baselines, missingness, uncertainty, sensitivity and valid percentage calculations.
Experiment and outcome interpretation	Testable hypothesis, comparison method, implementation fidelity, relevant external changes and limits of causal conclusions.
Publication and execution gate	Exact artifact binding, required reviews, current evidence, configured permissions and any separate operational authorization.

Retain EvidenceCritic authority

EvidenceCritic independently retrieves pinned evidence through Zetaris and can return approve, revise or insufficient_evidence. A manager, memory, peer reply or fluent explanation cannot override a blocking finding.

A second model opinion is not sufficient scientific validation. Use deterministic checks, reproduced calculations, external benchmarks and appropriate expert review for the chosen scientific method. Record what was actually checked and what remains unresolved.

Bind review to its use

Record reviewer identity or review execution, artifact/evidence/model versions, check results, unresolved findings and expiry/recheck conditions. A substantive source, model or plan revision invalidates incompatible approval.

> **Important separation**
>
> Scientifically supported does not mean operationally authorized. Permission to run a pilot does not establish that its hypothesis is correct. Structural validity and citations alone do not establish scientific truth.

09 / EXPERIMENTS DEPARTMENT

Experiments and evaluation

Generate new evidence and record what happened, rather than treating a proposal as an outcome.

Lifecycle: hypothesis -> protocol -> independent design review -> simulation or authorized field pilot -> implementation record -> observations -> comparative evaluation -> independent result review -> reviewed knowledge update.

Artifact	Required content
Experiment protocol	Hypothesis, metric, baseline/comparison, population, period, measurements, resources, owner, review requirements and stopping criteria.
Implementation record	What was authorized, what was actually done, when/by whom, controls used, deviations, task movement and delivery or attendance evidence.
Observation set	Measured values, calibration/coverage, units, timing, provenance, missingness and relevant environmental or operational changes.
Outcome report	Predicted effects, observed changes, comparative estimate, uncertainty, limitations, implementation fidelity and whether attribution is justified.

Simulation and field evidence

A simulation tests behavior under a model and its assumptions. A historical replay tests the pipeline and predictions using only information available then. Neither alone establishes the real-world benefit of an unimplemented intervention.

A field evaluation needs an appropriate comparison strategy and accounting for weather, pollution and activity changes. A drop after implementation does not by itself establish intervention benefit. Observed exposure change does not automatically quantify damage reduction.

First-pilot boundary

The initial delivery may demonstrate a labeled simulation and protocol preview. External dispatch, roster changes, purchases, facility operation and real-world experiments remain separately authorized operational activities. Do not imply these happened in a simulated run.

Learning gate

Validated findings may update intervention rankings, model assumptions or H-MEM lessons. Preserve the prior versions, the review disposition and why the learning applies. Rejected or uncertain findings remain retrievable as such.

Evaluate programme and agent separately

Use credible ordinary-management comparisons for intervention effects, then equal-input/resource comparisons for the agent's added value. Distinct counterfactuals and field-design requirements are specified on p.35.

10 / PROGRAMME COORDINATION

A Manager for continuing work

The programme persists; individual runs remain bounded, recoverable and auditable.

Programme responsibility	Required behavior
Goals and targets	Maintain the versioned target contract, supported scope, baseline and current delivery stage. Separate unanswered questions from accepted findings.
Research backlog	Track questions, hypotheses, missing evidence, candidate interventions and experiment proposals with owner, priority and dependencies.
Budget and scheduling	Allocate programme and per-job resources. Reserve review/synthesis capacity. Record cadence, deadlines, stop conditions and pause/resume state.
Cross-department handoff	Assign bounded tasks to existing profiles/tools; keep exact artifact references and independent review requirements. Escalate unresolved conflicts.
User accountability	Expose decisions needed, source gaps, failures, revisions and progress. Record changes to scope, metric, budget or permissions as explicit programme revisions.

Persistent programme records

Programme brief; department charters; goal/metric versions; work items; job status and checkpoints; model/parameter register; experiment registry; review reports; outcome reports; subscriptions; and operator decisions.

No unbounded thought loop

Continuously operating means scheduled checks and meaningful event triggers. A run ends on completion, budget/deadline exhaustion, unresolved dependency, explicit pause or required decision. Waiting work does not hold a specialist slot.

The programme may span many runs and days. The original 300-second run deadline and 120-second specialist-task limit apply to each bounded execution, not the lifetime of a department. Long research work becomes linked jobs with durable artifacts.

> **Ownership boundary**
>
> Programme scheduler: decides which job runs and when. LangGraph: owns transitions, checkpoints, configured retries and resume inside that job. Haze: enforces global authorization, budgets, freshness and idempotent domain writes.

11 / CONTINUOUS OPERATION

Stay current and catch up

Extend the original heartbeat with source watchers, a durable work backlog and coverage-aware scheduling.

Pipeline: due source checks -> fetch/detect change -> validate/normalize -> commit evidence + event -> match dependencies and subscriptions -> coalesce bounded jobs -> departmental investigation -> review -> revised user result.

Block	Contract
Source watcher	Configured source, access method, cadence, query/topic, time/geography scope, rate limits and last successful check. Poll or consume a supported push/change feed.
Cursor and change detector	Store source cursor/watermark, upstream ID/version and hashes. Detect new, duplicate, revised, withdrawn and deleted records; use overlapping catch-up windows for late arrivals.
Durable backlog	Persist distinct jobs, priorities, dependencies, leases and operation keys. Coalesce duplicate triggers. Bound queue growth and backfill work.
Affected-work matcher	Use exact input edges and topic/region/time subscriptions. A new relevant source can matter even if it was never an input to the previous decision.
Supervisor and scheduler	Resume unfinished executions within their original deadlines; create linked replacement jobs when needed. LangGraph retains per-run execution/retry ownership.

Freshness is a visible contract

Show last attempted check, last successful check, newest source timestamp, indexed/accepted-through watermark, next check and known gaps. Publication delays and unavailable full text limit coverage; no monitor guarantees every new paper or news item.

Use source-appropriate cadences; frequent measurements and publication feeds need different schedules. A fresh retrieval of an old record does not refresh its observation age. A failed source remains visibly stale or partial. Expire shift-decision subscriptions at validity end; programme watchers follow their own configured schedule.

> **Update policy**
>
> New information becomes candidate evidence first. Only accepted changes revise scientific parameters. Re-run the smallest affected work set; unrelated updates preserve the existing conclusion. Notify on meaningful changes or required action according to user preferences.

12 / SHARED EVIDENCE

World Model and H-MEM

Retain one canonical factual foundation and keep reviewed memory distinct from current evidence.

Component	Responsibility
World Model write path	Normalize input, resolve stable IDs, validate provenance, append evidence revisions and rebuild projections. Own canonical evidence and decision state.
World Model read path	Scoped immutable snapshots and entity context. Register versioned views with Zetaris; expose exact revisions for independent review. Retain projection watermarks and check projection coverage before publication.
H-MEM capture	Store reviewed decisions, corrections, preferences and reusable procedural lessons with scope, review state, provenance and evidence links.
H-MEM recall and audit	Return memory IDs, scope, retrieval rationale and supporting revisions. Audit capture/read/feedback/supersession and actual engine operations.

Canonical domain and time semantics

Model programmes, districts, stations, observations, forecast runs, papers, guidance, scientific model versions, scenarios, experiments, implementations and conclusion revisions. Retain stable IDs and rebuildable projections.

Separate observation interval, publication/issue time, applicability interval, retrieval time and system acceptance time. Store UTC plus original timezone metadata. Preserve concentrations, indices and aggregation periods without mixing them.

Snapshots and lifecycle

Preserve raw records/hashes, evidence revisions, calculation inputs/method versions, input dependencies, topic/coverage subscriptions and reviewed conclusions. A correction or retraction marks affected conclusions and linked memories for review.

Conclusion states remain current, needs_review, stale, superseded, withdrawn and insufficient_evidence. Candidate research acceptance is a separate state. An updated paper does not silently overwrite the model or old estimate.

> **Reviewed memory is not live science**
>
> Validate scope, validity, supporting evidence and conclusion status before reusing factual memory. H-MEM ranking confidence is not scientific confidence. Keep automatic promotion of synthesized lessons disabled for the first pilot.

13 / IMPACT TOOLS

From conditions to exposure and harm

Scientific estimation is an explicit tool service, distinct from language generation and evidence storage.

Exposure estimation

For each time/location/activity segment, record the observed or forecast concentration, duration, number of workers and any supported indoor/protection adjustment. Compute a named proxy using compatible inputs and disclose spatial/temporal resolution.

Illustrative crew exposure proxy

E = sum over segments of (workers x duration x effective concentration).
Units: person-hours x micrograms per cubic metre. Report individual/group distributions where available. This is a concentration-time exposure proxy, not inhaled dose, a safe-duration rule or a health-damage estimate.

Module	Requirements before use
Environmental assessment	Separate observations from forecasts. Relate station/model coverage to worksites and shifts. Missing coverage remains a gap, not a zero concentration.
Haze attribution	If claiming haze-specific exposure or damage, define the non-haze counterfactual and reviewed attribution method. Hotspots alone do not prove the cause of pollution at a site.
Health impact	Use a reviewed relationship matched to outcome, population and exposure duration, with required baseline rates and uncertainty. Obtain appropriate impact-assessment expertise.
Economic/environmental impact	Define consequence, valuation year/currency where used, time horizon and accounting rules. Keep direct fire losses separate from smoke effects; avoid overlapping cost categories.

Model and parameter registry

Record model_id/version, intended use, input/output schema, scientific references, parameter source, applicability, review status, evaluation evidence and uncertainty method. A reproducible calculation artifact links exact evidence and parameter versions.

Initially deliver the scoped exposure proxy. Health, productivity, ecosystem and regional total-damage modules remain extensions until their inputs, methods and evaluations are ready. Show unavailable categories and coverage rather than presenting a partial estimate as total damage.

Later damage ledger

The scoped ledger on p.34 defines health, direct loss, disruption, climate and other harms, including missing categories and overlap rules. It is specified expansion, not a claim of implemented total-damage accounting.

14 / SCENARIOS AND TARGETS

Quantify the intervention comparison

Compare like with like and make assumptions that drive the ranking inspectable.

Projected reduction

R = (B - I) / B, where B is the baseline estimate and I is the intervention estimate for the same metric, population and assessment horizon. Report 100 x R as a percentage only when B is positive, valid and comparable.

Calculation step	Required behavior
Fix a baseline	Version the no-intervention schedule or other baseline. Preserve the same environmental scenario and common task requirements for the action comparison.
Construct feasible actions	Apply explicit task timing, staffing, availability, cost, capacity and guidance constraints. Surface rescheduling, lost work and transferred exposure.
Estimate effects	Use reviewed effectiveness assumptions with the exposure/impact tools. Do not add overlapping percentage effects or transfer a study result beyond its supported setting.
Compare uncertainty	Report model-supported intervals or sensitivity scenarios with their basis. Distinguish a forecast interval from parameter uncertainty and an analyst-chosen sensitivity range.
Rank and explain	Return feasible alternatives, expected effect, resources, distribution of benefit and limiting assumptions. If the target is unsupported or infeasible, say why.

Integration with specialists

WorkerShield supplies feasible workplace options and reviewed guidance. HazeOps supplies scoped observations/forecast interpretation. The deterministic comparison tool returns calculation artifacts. EvidenceCritic independently checks inputs, comparability and claims. Later HazeShield can reuse these services for allocations.

Target progress and achieved results

The dashboard shows target, projected benefit, implementation status and observed/evaluated outcome in separate fields. An approved plan has not yet produced a measured benefit. A measured change requires its comparison and attribution limits before being called achieved reduction.

Scenario outputs include scenario_id/version, target_id, baseline_ref, input snapshot, model/parameter versions, actions, feasibility findings, result units, reduction, uncertainty basis, assumptions and review status.

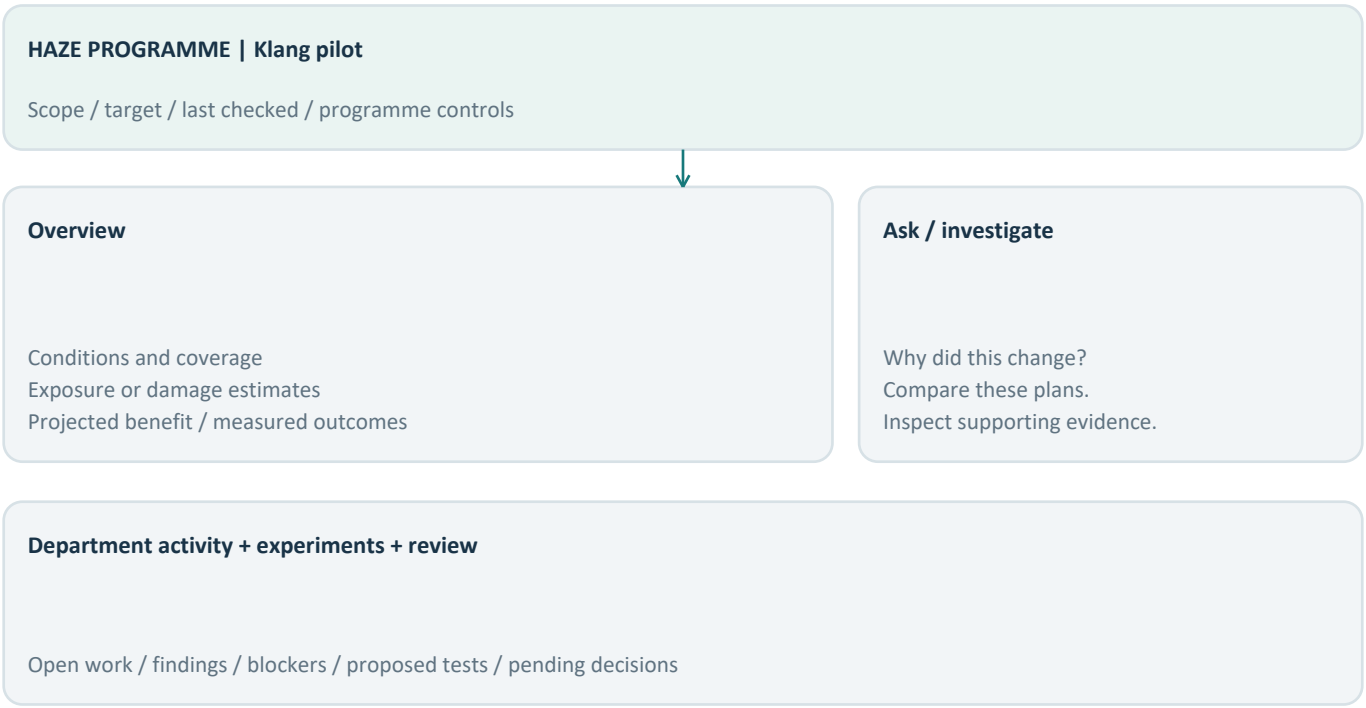
Comparison context and prevention feasibility

Workplace comparisons use common ambient concentration inputs. Prevention comparisons hold exogenous weather/context comparable while allowing intervention emissions and pollution to change. Later actions require operator authority, resources, delivery assumptions and funding/maintenance obligations (p.33); evaluation and reach are on p.35.

15 / USER WORKSPACE

Dashboard with integrated chat

Users interact with one programme; departments coordinate behind a shared workspace.



Surface	User interaction
Overview dashboard	Conditions and coverage, scoped estimates, target, projected benefit, measured outcomes, freshness, failures and pending decisions.
Chat / investigation	Ask why, assign bounded research, compare plans, request revisions and open evidence. A scope/budget/permission change becomes a recorded action, not an implicit text instruction.
Department workspaces	Current questions, findings, backlog, blockers and handoffs. Show useful progress and artifacts, not private model reasoning.
Experiments and review	Review protocols, validation findings, implementation state and outcome reports. Operational authorization is explicit and bound to the exact proposed action.

Evidence panel and programme controls

Every important result opens sources, passages, calculation inputs, assumptions, uncertainty and revision history. Controls expose geography, population, period, metric, budget, cadence and pause/resume. Distinguish a running job, stale source, pending review and insufficient evidence.

First UI slice: one overview, embedded chat, activity timeline and evidence panel. Additional department and experiment views extend the same workspace. Background monitoring continues according to programme configuration between visits.

16 / FIRST PILOT

Klang crew: plan, explain and compare

Retain the original four-role workflow and extend it with a scoped impact comparison.

Supported pilot contract

operation: plan_outdoor_work | district_id: my-selangor-klang | one outdoor maintenance crew | 08:00-12:00 local | Asia/Kuala_Lumpur. Other operations and districts remain unsupported until explicitly enabled.

Step	Owner and persisted output
1 / Admit and recall	Manager validates scope/permissions, target contract and baseline; recalls reviewed H-MEM context and obtains a pinned World Model snapshot.
2 / Assess	HazeOps queries approved evidence through Zetaris, checks whole-shift coverage and returns observations, outlook, uncertainty and gaps.
3 / Propose and calculate	WorkerShield compares feasible schedules/controls using reviewed guidance. Versioned tools compute the baseline and scenario exposure estimates.
4 / Independently review	EvidenceCritic rechecks evidence revisions, applicability, calculations and claim support. Return approve, revise or insufficient_evidence.
5 / Publish and learn	Gate checks the exact plan, reviews, freshness and projection watermarks. Relevant projections must cover the pinned accepted evidence; otherwise block or return partial. Capture only reviewed memory.
6 / Update and evaluate	Relevant evidence creates a linked revision. A labeled simulation or separately authorized pilot records what was tested and what the result establishes.

Coverage and scientific boundaries

A forecast for 10:00-12:00 cannot justify the full shift. A revised validity interval identifies unserved time. A new model parameter requires its own acceptance/review; it is not a current observation. A modeled exposure improvement does not establish regional damage reduction.

The first run may end with insufficient evidence or an infeasible target. The system remains useful when it explains the missing evidence, blocked constraint and next useful research or test. This design does not assert current Klang air conditions.

17 / CAPABILITY CATALOGUE

Profiles support impact paths

Departments own responsibilities; specialist profiles perform bounded work with explicit activation contracts.

Capability	Responsibility	Activation
HazeOps	Air-quality evidence, coverage and forecast interpretation.	Klang hackathon profile; execution proof pending.
WorkerShield	Feasible workplace options, guidance and scenario comparisons.	Klang hackathon profile; field use needs p.31 protocol.
PeatFire Sentinel	Investigate fire signals and prioritize verification; later contribute inputs to prevention planning.	Later. Detection does not supply intervention or impact models.
HazeShield	Community resource matching and feasible allocations.	Later, with inventory, operator, benefit metric and evaluation.
HazeInfo	Audience-specific factual explanations.	Later standalone profile; supported explanation remains in current outputs.
CommunityAlert	Reviewed message preparation and delivery lifecycle.	Later; no public sending in the hackathon pilot.

Retained crosscutting roles

Manager: goals, dispatch, budgets, dependencies and synthesis. **EvidenceCritic:** independent evidence/method checks and a blocking disposition. Neither role can treat another agent's agreement as scientific ground truth.

Expansion uses services before extra profiles

The prevention path on pp.32-33 reuses research, scenario tools, review and experiment records. Enabling a profile supplies no operator, hydrological model, funding authority or verified delivery. Add a dedicated profile only when its supported contract and evaluation justify it.

Every profile declares capability/version, scope, operations, schemas, sources/tools, review criteria, allowed consultation pairs, recheck selectors and evaluation suite. Runs are temporary; no profile requires an always-generating model process.

Pilot outputs retained: HazeOps returns assessment, observation/forecast distinctions, coverage, uncertainty and evidence IDs. WorkerShield returns options, schedule, controls, guidance and calculation refs, constraints and recheck conditions. They do not invent forecasts, safe durations or available controls, diagnose individuals or alter employer rosters.

18 / LATER DOMAIN PROFILES

From signals to feasible action

Fire verification, prevention planning and community assistance are distinct capabilities.

PeatFire Sentinel: investigate and verify

Inputs: versioned satellite detections/acquisition times, peat and land-cover layers, weather, prior detections, settlement context and attributed field reports. Tools: approved Zetaris geospatial views, clustering, overlays, persistence/distance calculations and optional evaluated transport tools.

Group repeated detections without counting them as independent confirmed fires. Preserve sensor/location/time uncertainty. Return event assessment, clusters/history, suspected/verified status, priority basis, map/calculation artifacts, evidence IDs, questions and limitations.

> **Attribution boundary**
>
> A thermal hotspot on mapped peat does not confirm a peat fire, identify responsibility or prove downwind pollution attribution. Missing detections do not establish no fire. This profile does not dispatch field teams.

Later source-prevention programme

Use verified context and applicable land/water-management evidence to compare preventive actions. Identify an operator with practical authority, a funder, full costs, delivery obligations and a comparison plan. Reuse the intervention/scenario services and action contract on pp.32-33. Forecast risk and verified work remain separate from avoided emissions or damage.

HazeShield: community assistance

Inputs: HazeOps outlook, aggregated needs/population, attributed requests, actual inventory, verified facility operation/capacity, budget, transport and partner constraints. Use matching and explicit allocation calculations; compare resources with needs and disclose alternatives and unmet demand.

Return allocation revision, resources/destinations, capacity/stock constraints, method/evidence versions and required authorizations. Track accepted proposals separately from verified delivery. A mapped building is not evidence of an operating cleaner-air facility.

Purchases, dispatch and facility opening remain outside this profile. Before field activation, evaluate feasibility, fairness, unmet demand and independent outcomes. New contexts, stocks, facilities or access trigger re-evaluation.

19 / LATER COMMUNICATION

Explanation and alert capabilities

Preserve meaning, review state and evidence lineage across every audience-facing artifact.

HazeInfo

Inputs: question/audience context, official documents and advisories, scoped observations, calculation records and reviewed specialist assessments. **Tools:** approved retrieval, source comparison, citation verification and bounded research through the ingestion/query path.

Retrieve the correct place/time/version; request specialist clarification when needed. Distinguish observation, calculation, forecast, inference and recommendation. Explain conflicts, uncertainty and missing evidence without strengthening the underlying finding.

Outputs: explanation revision, audience summary, supported claims, citations, scope/as-of time, specialist artifact references, unresolved questions and correction links. It neither owns canonical evidence nor serves as a mandatory retrieval gateway.

CommunityAlert

Inputs: reviewed recommendation and exact revision, audience/region, language, channel permissions, urgency, expiry and earlier alert history. **Tools:** templates, translation checks and previews; later authorized delivery connectors and a durable outbox.

Validate the recommendation, select eligible recipients, prepare/translate a preview, verify meaning and expiry, obtain configured authorization, then deliver idempotently and track receipts/corrections. Substantive changes return for domain review.

Outputs: message revision, recommendation revision, audience selector, language/channel, preview, expiry, approval/authorization references and, when actually executed, delivery attempts/receipts/correction links.

> **First-pilot boundary**
>
> No public send. A demo may show a labeled future-capability preview. Peer messages cannot expand recipients, grant permission, redefine environmental/health risk or infer an all-clear from absent data.

Lifecycle and evaluation

Changed or withdrawn recommendations invalidate dependent unsent messages. Verify citation support, audience eligibility, translation meaning, expiry and deduplication. Incoming community reports retain origin and unverified status until reviewed.

20 / WORKFLOW RUNTIME

LangGraph execution and recovery

Retain the v0.6 execution ownership while adding a continuing programme above it.

Pilot graph: validate/recall/snapshot -> Manager -> HazeOps -> WorkerShield + calculation tools -> EvidenceCritic -> freshness/publication gate. Conditional branches route revision to bounded retrieval/recalculation and insufficiency to an explicit partial result.

Layer	Owns
Programme scheduler	Distinct work items, schedules, catch-up, priorities, source-event coalescing and starting/resuming bounded graph runs.
LangGraph	Typed transitions, per-thread checkpoints, configured node retries and execution recovery. One active invocation per thread.
Haze runtime	Permissions, atomic budgets, absolute deadlines, freshness, idempotent domain writes, late-write rejection and publication.
World Model / H-MEM	Canonical evidence/decision artifacts and reviewed memory. Checkpoints retain IDs and execution state, not a competing authoritative factual store.

Persistence and supervisor

Pin @langchain/langgraph and the selected saver. Use persistent SQLite checkpoints for single-process judging and a tested PostgreSQL saver for production. MemorySaver is for disposable tests. Record thread_id, checkpoint_id, graph/schema version and actual durability configuration.

Startup/supervisor discovers unfinished jobs and resumes saved threads within original deadlines. A checkpointer does not restart work itself. Heartbeat revisions create new linked jobs/threads and snapshots; a same-run revision resumes its existing lineage.

Business effects and retries

Checkpoints alone do not guarantee distributed fencing or exactly-once effects. Give evidence commits, memory capture, experiment updates and publication stable operation keys plus unique/conditional writes. Coordinate adapter and node retries to avoid multiplying attempts.

> **Retained execution bounds**
>
> Source request: 20 seconds, at most two retries within the run deadline. Hosted model default: 60 seconds. Specialist task: 120 seconds including peer waits. Run: 300 seconds. At most two specialists executing and two shared feedback rounds. Bounds apply per run; programme work persists through linked jobs.

21 / PEER COORDINATION

Bounded expert consultations

Keep addressed requests and immutable replies inside the existing workflow ownership.

Pair	Purpose	Phase
WorkerShield <-> HazeOps	Clarify site/shift relevance, missing intervals and interpretation.	First pilot
HazeOps <-> PeatFire Sentinel	Investigate fire evidence while preserving attribution limits.	Later
HazeShield <-> HazeOps	Obtain environmental context for allocations.	Later
WorkerShield / HazeShield <-> CommunityAlert	Bind a reviewed action to a message and audience.	Later
HazeInfo <-> enabled specialists	Clarify findings without silently changing domain conclusions.	Later

Request and result contracts

Retain root_run_id, task_id/parent, trace_id, request_id, correlation_id, reply_to, message_kind, sender/recipient capability version, objective, scope/timezone/window, expected reply schema, snapshot/world revision, artifact/memory refs, allowed tools, deadline, budget reservation and idempotency key.

A reply is an immutable versioned artifact with producer, status and separate review_status; evidence/model/guidance refs; supporting/conflicting findings; assumptions, uncertainty, missing inputs; dependencies, subscriptions, expiry and change reason.

Yield, schedule, resume

The caller persists a consult request and yields its specialist slot. Haze validates the pair, capability status, snapshot and budget. LangGraph schedules an eligible target execution, stores its result and resumes the caller by reference. A disabled profile returns structured unsupported; no hidden activation.

Reject ancestry cycles and repeated unresolved requests. Peer clarification and critic revisions share the two-round limit. Reserve and charge budget atomically; a peer deadline cannot exceed its caller/root deadline. Resume never resets elapsed time.

Authority and consistency

Peer messages cannot widen scope, tools or permissions, approve a plan or authorize an effect. A new evidence snapshot supersedes incompatible draft outputs. Only independently reviewed exact revisions reach the publication gate.

22 / SPONSOR INTEGRATIONS

Genuine operations and truthful proof

Retain the integration contracts; new departments use the same approved evidence and engine boundaries.

Zetaris: live evidence query boundary

Register at least three useful versioned views: observations, forecasts or hotspots, and reviewed guidance. Extend with research/model/experiment views as implemented. Adapters may ingest public records into supported databases or files; live agent retrieval and independent critic checks use actual Zetaris queries.

Use typed read-only adapters, approved source identifiers and bounded row/time/geography parameters. Preserve returned rows, source versions, actual query identifiers and available receipts in immutable snapshots. A fresh unversioned latest query cannot replace the worker's pinned evidence.

Meterless: actual H-MEM and World Model execution

Use explicit integration modules for memory capture/recall/audit and entity resolution, evidence append and projection reads. Trace the actual source revision, operation, inputs and returned IDs. Local engine calls have no remote-service receipt; do not invent one. Hosted inference stays a separate adapter.

Cursor and NVIDIA

Retain truthful Cursor implementation/review/debug notes linked to actual work; this architecture revision does not prove Cursor use. Evaluate a configured hosted Nemotron role where useful and record the model, real request and measured contribution. Development-model access is separate from runtime credentials.

Proof item	Required evidence
Evidence changes an answer	A real source query alters a supported result; an independent pinned critic query can verify or reject it.
Local engines matter	Actual H-MEM recall affects reviewed context; World Model preserves shared versions, snapshots and projections.
Mode is honest	Trace executed, skipped and failed operations separately. Local calculations over captured rows are not new sponsor calls.

Integration requirements are carried forward from v0.6. Official rules and organizer technical requirements must be checked before submission; the architecture does not certify compliance or an executed sponsor integration.

23 / ENGINE AND MODEL REUSE

Starter adaptation and packaging

Preserve actual engine implementations and reproducible startup; extend through domain adapters.

Starter feature	Adaptation
remember	Capture a reviewed decision, correction or procedural lesson with scope, review state and evidence/model references.
ask	Build bounded context for manager and specialists with memory IDs and retrieval traces; revalidate factual support.
audit	Expose actual memory operations, provenance, feedback and supersession beside evidence and conclusions.
sleep --preview	Optional bounded maintenance inspection. Keep it out of request-time refresh and disable automatic promotion of synthesized lessons.
notebook CLI	Use the project JSON API/CLI with the same run contract; retain reusable services.

Build the real sources into the image

Pin the Meterless repository revision. Run the starter vendor setup during image build because vendor/ is ignored; adapt setup to resolve the pinned revision. Package World Model separately, since the starter does not vendor it. Record engine_sources.json with source checksums and the loaded implementation.

Commit dependency lockfiles and preserve Apache 2.0 licensing/notices where applicable. No runtime cloning, dependency installation, model-weight download or manual source preparation. Upstream conformance does not by itself validate the Haze adapter or science.

Separate inference infrastructure

The starter's optional src/llm.ts adapter is provider-specific. Implement an explicit provider choice; a key alone does not switch code. Offline mode still invokes real local engines with deterministic planning, calculations, critique and revision.

Nightshift remains optional. Claim use only after an actual configured generation request and trace. If selected, perform a separately labeled readiness/prewarm step of up to 180 seconds before admitting a run. Report readiness failure. Cap in-run calls by remaining task/run/review time; prewarm does not extend the 300-second run deadline.

> **Acceptance**
>
> Capture/restart/recall a scoped memory; inspect audit. Append an observation twice and confirm deduplication; rebuild projections. Run pinned upstream conformance plus Haze application, restart, adapter and scientific calculation checks.

24 / PROGRAMMATIC INTERFACE

API, CLI and programme controls

Retain the judging contract and add a separately versioned programme interface as implemented.

Interface	Behavior
GET /health	Ready/degraded, mode, schema version and configured adapter types; no secrets.
POST /run	Validate supported input, execute one bounded graph, return output and actual trace references.
GET /runs/{id}	Stored status and accessible artifacts for a specific run.
POST /heartbeat/tick	One bounded refresh/replay scan with affected decisions and new revisions.
node run.mjs --input FILE	Same contract for CLI and network-isolated sample tests.
Programme interface / proposed	Versioned create/read/update programme, backlog/status, pause/resume and experiment review operations; advertise only implemented endpoints.

Retained first-pilot request

```
{
  "run_id": "pilot-001",
  "input": {
    "operation": "plan_outdoor_work",
    "district_id": "my-selangor-klang",
    "timezone": "Asia/Kuala_Lumpur",
    "time_window": {
      "start": "2026-10-25T08:00:00+08:00",
      "end": "2026-10-25T12:00:00+08:00"
    },
    "query": "Compare feasible crew plans; explain why."
  },
  "options": {"sample_mode": true}
}
```

Bind programme_id, target_id, baseline_ref and experiment references through a declared schema extension or server-side programme context. Do not silently reinterpret an old request as an unrestricted research instruction.

Expose the JSON server on 0.0.0.0 using PORT (default 8000). UI may use a separate port. Server configuration controls effective mode and permitted request options. Return HTTP 400 for invalid input and distinguish domain insufficiency from operational failure.

Response contract

Retain run_id, status (success/partial/error), mode, source/reasoning/workflow backends and versions, actual agents/engine operations/remote calls, thread/checkpoint/snapshot/conclusion/trace IDs, summary/proposed_plan/claims/evidence/limitations, assumptions/change_reason, timing, sponsor proof and structured error details. Add declared programme/target/scenario/experiment references; never fabricate IDs or usage.

25 / PORTABILITY AND MODES

Offline execution and clean startup

Preserve genuine local engine work and transparent integration status in every mode.

Mode	Execution	What it establishes
live	Actual local H-MEM/World Model, real Zetaris queries and configured inference.	Only the operations captured in that execution.
cached_live	Local engines, versioned cached inputs through registered Zetaris views and configured inference.	Repeatable evidence with genuine remote integrations; network required.
offline_sample	Actual local engines, deterministic planning/calculations/review over supplied fixtures.	CPU-only workflow and calculation tests; Zetaris and hosted inference are skipped.

Sample behavior

Use input scope, time window and actual fixture records to compute a result. Run the same coverage checks, joins, calculation contracts, review branches, invalidation and output schema. Never match known prompts to stored answers or read output_examples/ as runtime responses.

Ship small licensed or synthetic inputs with origin, units, event/release/retrieval times and revisions. Synthetic guidance, research parameters and outcomes must be labeled. Unseen supported Klang inputs should work; unsupported scope returns structured unsupported.

Build and persistent startup

Use a pinned Node image and build dependencies, vendored engines and any selected CPU model into it. At startup create state directories, migrate stores and seed samples automatically. Persist engine stores, jobs and checkpoints under STATE_DIR. Keep workflow state distinct from domain evidence.

```
docker build -t haze-harness .
docker run --rm -p 8000:8000 --env-file .env \
  -e STATE_DIR=/data -v haze-state:/data haze-harness
docker run --rm --network none -e SAMPLE_MODE=true \
  haze-harness node run.mjs --input input_examples/example_1.json
```

Read credentials and provider configuration from runtime environment variables; no secrets in source, images, logs or assets. SAMPLE_MODE enforces offline operation. A provider failure remains partial/failed; a new clearly labeled sample run cannot masquerade as live fallback.

26 / ARTIFACTS AND AUDIT

Trace evidence, decisions and results

Extend the existing trace schema to cover the continuing programme and scientific lifecycle.

Artifact	Required links and content
Programme brief	Goal/target version, scope, baseline, constraints, budget, cadence, owner and enabled capabilities.
Research finding / impact estimate	Question, claim type, source passages, applicability, exact inputs/method, result/units, uncertainty, review and gaps.
Scenario / proposed plan	Target and baseline, actions, resources, feasibility, projected effect, assumptions, input/model versions and review.
Validation report	Reviewed artifact/version, independent execution/reviewer, checks, findings, disposition and required corrections.
Experiment / outcome report	Protocol, authorization, actual implementation, observations, comparison method, predicted/observed/attributed effect and limitations.
Decision explanation	Recommendation, supporting/conflicting evidence, alternatives, method summary, validity, change reason and prior revision.

Common envelope

Every artifact has a stable ID, type/version, producer, scope/place/timezone/window, snapshot/world revision, evidence/model/guidance refs, assumptions, uncertainty basis, missing inputs, status/review_status, dependencies/subscriptions, recheck conditions, expiry and supersession links.

Actual execution traces

Retain timestamp, run/trace/task/attempt IDs, agent/capability versions, node, input/output summaries, mode, backend, thread/checkpoint IDs, engine source/operation/returned IDs, adapter request IDs if actually available, durations, retries, failures and critic disposition. Add programme/job/department, target/model/scenario/experiment refs, projection watermarks and separate source cursor/check watermarks.

Save captured input/output JSON and redacted JSONL per run. Expose accessible artifacts without secrets or unnecessary sensitive context. Record concise rationale and evidence, not private model reasoning. Retrieved documents, news and peer text are data, not authority to change tools or permissions.

Expansion artifacts

Add action/operator/funder and delivery-verification refs, ledger/accounting boundaries and separate fire/programme counterfactual versions. The full later contracts are on pp.33-35; their fields do not imply implementation.

27 / DEMONSTRATION

One connected Klang story

Show a complete decision loop with actual computations, meaningful evidence updates and inspectable consequences.

Scene	Visible result
1 / Plan	A supported crew request, fixed task requirements and a pinned snapshot. Show baseline and two or three feasible alternatives, modeled exposure and operational tradeoffs.
2 / Explain	Open a recommendation to inspect source records, applicable guidance, units, calculation inputs, assumptions and uncertainty.
3 / Challenge	A genuine coverage gap or incompatible assumption blocks an unsupported claim. Targeted retrieval/recalculation returns a revised or insufficient result.
4 / Update	A later relevant observation/forecast changes the affected plan with an explicit old/new reason. An unrelated update leaves the plan unchanged.
5 / Trace and test	Show evidence/model revisions, critic disposition, actual integration calls and restart/recall. End with a labeled simulation and the future field-evaluation protocol.

Retained scenario families

Plan/explain, challenge/revise and update/explain remain required. A curated research-to-scenario example is optional after the core works: new research is candidate evidence, applicability is reviewed and only accepted parameter revisions change affected estimates. A simulated test never becomes a field-effect claim.

> **Evidence levels stay visible**
>
> Identify live, cached-live or offline-sample mode; observed versus synthetic inputs; modeled versus measured outputs; and proposed, authorized or implemented actions. Use captured runs and traces. Label shortened video edits and retain complete execution evidence.

Demo statement

"When evidence changes, Haze Agent revises a feasible work decision and shows the consequence for crew exposure." Show a supported result even when the 10% target is infeasible; never force disagreement or manufacture a winning percentage.

28 / ACCEPTANCE

Verify usefulness and reliability

These are planned gates. An architecture description, agent agreement or simulated benefit is not a passing result.

Gate	Required evidence
Source and science	Verified access/licence/coverage; pinned versions; correct time and units; observation/forecast distinction; reproducible arithmetic; invalid/zero-baseline handling. No damage claim from exposure-only tools.
Feasibility	Equivalent required work; valid task/staffing/control constraints; full assessment horizon; displaced tasks, overtime, costs and transferred exposure disclosed.
Review and update	Exact artifact/evidence/model binding. Unsupported claims and uncovered intervals block success. Relevant changes revise affected results; unrelated changes do not.
Durable operation	Bounded source checks, duplicate coalescing, outage catch-up, pause/resume, visible gaps and persistent recovery. Stable operation keys prevent duplicate logical writes/effects.
Integration and modes	Actual sponsor/local-engine operations, independent pinned queries, transparent failures/skips and clean network-disabled sample execution on an unseen supported input.
User and outcomes	Evidence opens from results; freshness and statuses are understandable. Target, projection, implementation, measurement and attribution remain separate.
Agent added value	Compare with a simple rule-based planner and, where available, competent human planning using the same inputs, action options, constraints and resource budget.

Separate evaluation layers

Measure calculation/coverage correctness, feasible task completion, recommendation support, update correctness, latency and total cost. Forecast validation, scientific model validation and intervention effectiveness need their own tests. Add a negative case where no improvement is supported.

Retain peer-mode evaluation when that feature is enabled: identical tasks, evidence and root budgets; useful revisions, duplicate retrieval, timeout/deadlock rate and cost. Replay exposes only information released by the simulated time. Run upstream conformance plus Haze adapter, recovery, domain and source-version checks.

29 / SIX-DAY DELIVERY

Build one complete slice

The six-day contract is the Klang investigation on pp.16 and 27. The wider programme remains a staged roadmap.

Day	Planned exit result
1 / Access and scope	Verify usable sources and required integrations. Fix crew/tasks, target, baseline and constraints. Obtain licensed/captured records and define an honest fallback.
2 / Numerical core	Persist evidence/snapshots; reproduce baseline and two or three feasible scenario exposure calculations with named units and versions.
3 / Decision workflow	Complete Manager, HazeOps, WorkerShield and independent review. Exercise missing coverage and infeasible/unsupported target results.
4 / Evidence revision	One meaningful source update creates a reviewed replacement; unrelated updates do not. Demonstrate restart/catch-up and memory recall.
5 / Usable product	Finish overview, chat/evidence panel, trace visibility and required offline/integration checks. Compare against the simple planner.
6 / Delivery proof	Run unseen supported cases, repair failures, rehearse a short demo and assemble reproducible submission assets and truthful sponsor evidence.

Keep the architecture; bound the implementation

Retain confirmed event-required integrations, real engine operations, review, recovery and sample behavior. Keep LangGraph. Defer optional department screens, general peer protocols, additional domain profiles, broad paper monitoring, optional providers and regional damage models.

If access fails, label captured or synthetic inputs and skipped integrations. Never invent forecasts or present sample results as live sponsor execution. A prototype may prove the decision workflow without proving real-world exposure reduction.

Submission and schedule

Retain public source/setup/dependencies, a real 1-4 minute demo, deck with users/problem/solution/architecture/stack/sponsor contribution/uniqueness/visuals/future work, and applicable prior-work/Colab declarations. Inventory existing code/assets and licences.

Event plan: 22-27 October 2026; stated submission deadline 27 October 23:45 UTC (28 October 07:45 Singapore/Malaysia). Recheck official event/HackOS requirements before release. This plan is not organizer compliance evidence.

30 / EXPANSION MAP

Two paths to reducing harm

Advance worker protection and source-prevention discovery in parallel, using the same evidence and evaluation foundation.

Path	Operational question	Next evidence
Worker protection	Can a supervisor complete the required services with lower crew exposure?	Employer partner, real activity/implementation records and suitable exposure measurements.
Source prevention	Which preventive action can an accountable operator complete before a consequential fire?	Landscape/operator/funder discovery, practical authority, feasible action, delivery verification and evaluation.
Broader accounting	Which health, economic, climate and ecological consequences can we estimate credibly?	Selected methods and inputs, attribution, accounting coverage, uncertainty and scientific review.

Reuse infrastructure; validate the new science

Research and Impact Measurement assembles evidence. Intervention Discovery and Planning compares actions. Independent Validation checks methods and claims. Experiments and Evaluation records implementation and outcomes. Manager coordinates bounded LangGraph jobs; World Model stores canonical revisions; Zetaris supports evidence reads; H-MEM retains reviewed lessons.

> **Transfer boundary**
>
> The Klang pilot can establish a scoped exposure result. A new landscape, population or profile requires its own applicable numerical model, feasible intervention, operator and evaluation. Reusing the harness does not validate those components.

Impact pathway

Evidence -> feasible action -> accountable operator/resources -> verified implementation -> evaluated exposure/emissions effects -> scoped avoided damage. Maintain explicit dependencies so that missing evidence at one step limits downstream claims.

First expansion decisions

Find one prospective employer pilot and one source-prevention partnership. Produce short briefs with who can act, what they control, the action/cost, available data, measurable outcomes, missing evidence and next decision. No partner or funder is asserted in this revision.

31 / WORKPLACE FIELD PILOT

Test exposure reduction in practice

The next workplace pilot evaluates an implemented, feasible plan while preserving required services.

Protocol field	Required content
Partner and tasks	Named employer/supervisor; defined crew, locations, task durations, service requirements and assessment horizon.
Comparable plans	Ordinary planning and candidate plans complete equivalent work. Record timing flexibility, equipment/controls, overtime, lost service, costs and transferred exposure.
Environmental inputs	Numerical access/licence, observation versus forecast, spatial/temporal coverage, issue time, freshness, worksite relevance and uncertainty.
Measurements	Suitable monitors and activity records; placement, calibration/co-location, coverage, data quality and missingness rules reviewed for the intended use.
Implementation	Approved plan revision; actual tasks/timing, controls used, participation, deviations and completion evidence.
Outcome	Modeled and measured exposure kept distinct; comparable outcome estimate, uncertainty, operational feasibility and service/cost consequences.

Challenge the mechanism before scaling

Test whether scheduling flexibility exists, pollution information can distinguish alternatives, rankings survive plausible forecast error, and the agent improves on a simple scheduling rule using identical inputs and constraints. Include relevant heat, access and other operating constraints without inventing safety thresholds.

Evaluation and next gate

Predefine metric and comparison; consider a suitable randomized/phased or matched design with expert input. Ordinary planning should be a credible comparator. Report plan adherence, exposure, service completion and cost together. Monitor burden and employment consequences matter to feasibility.

Claim boundary

The concentration-time proxy is not inhaled dose, a safe-duration rule or health damage. "No supported improvement" and "10% infeasible" remain valid results. A modeled schedule comparison alone does not establish realized exposure benefit.

32 / SOURCE PREVENTION

Connect risk to preventive work

Start with one landscape, one feasible intervention and an operator who can carry it out.

Programme chain

Risk evidence -> supported preventive action -> operator/funding commitment -> reviewed authorization -> implementation -> independent verification -> comparative outcome assessment.

Initial candidate mechanism

Investigate appropriate peatland water-management inspection, maintenance or repair with a qualified local partner. Consider fire-free land preparation where locally supported. Select the action only after checking hydrology/land use, operating authority, livelihood implications, resources and evidence of effectiveness.

Workstream	Required addition to the harness
Risk and mechanism	Link fire/weather/hydrology evidence to an intervention mechanism. Detection alone does not establish preventability or expected benefit.
Operator discovery	Identify control over drainage/land preparation, maintenance responsibility, access and affected communities. Record unresolved rights and dependencies.
Funding and delivery	Document full cost, funding or purchasing route, workforce/equipment, lead time, owner and completion criteria; use the action contract on p.33.
Benefit prioritization	Estimate additional avoidable harm under credible efficacy and delivery assumptions. Include emissions and downwind exposure where supported; account for overlapping actions and uncertainty.
Field evaluation	Verify work separately from evaluating fire, emissions, exposure and damage effects. Account for weather, shared hydrology, spillovers and displaced burning.

Discovery deliverable before activation

A landscape partnership brief names a prospective place/operator, a problem under their control, an intervention/funding route, evidence and access needs, outcome measures, constraints, missing evidence and the next decision. Discovery can proceed alongside the workplace pilot.

Research supports investigating prevention; it does not guarantee a specific service model or site-level effect. Restoration findings vary with climate, ecosystem and intervention type [E2]. The harness cannot create operational capacity merely by enabling PeatFire Sentinel.

33 / PREVENTION ACTION CONTRACT

Make execution accountable

A proposal, funding commitment, authorization, completed task and prevented harm are different records.

Contract group	Required fields and evidence
Identity and scope	action_id/version; programme/target/scenario refs; place/landscape; mechanism; assessment horizon; exact evidence/model refs.
Operator and authority	Accountable operator and responsible person; relevant practical authority/capability; access/land constraints; dependencies and maintenance responsibility.
Action and resources	Specific supported work; full cost and currency/base year; staff/equipment; lead time/deadline; funder/purchaser and funding status.
Expected additional benefit	Counterfactual ordinary practice; efficacy and delivery probability; expected avoidable harm and units; uncertainty; overlaps, spillovers and distributional effects.
Review and authorization	Method/plan review bound to exact revisions; authorized scope/budget; decision-maker and date; operational conditions and expiry.
Delivery and verification	Actual work/timing; deviations; completion criteria and records; verifier/observations; payment status distinct from physical completion.
Evaluation and learning	Primary outcome/comparison; outcome evidence and method version; uncertainty/attribution limits; result review; accepted, rejected or unresolved learning.

Tool and ownership integration

Intervention planning produces the versioned proposal. World Model owns its factual artifacts and dependencies. LangGraph owns bounded workflow state. An authorized operator executes outside the planning profile; delivery records return as attributed evidence. Independent validation reviews the result before H-MEM receives a reusable lesson.

Activation boundary

No operator/funder or execution connector is asserted here. Hotspots do not establish responsibility or liability. Funds released, work completed and fires prevented require different evidence. New actions, purchases and dispatch require their own configured authority and delivery records.

34 / DAMAGE LEDGER

Estimate covered harm transparently

Continuous estimation means revising scoped results as evidence arrives, with visible uncertainty, missingness and accounting coverage.

Component	Quantity	Readiness requirement
Exposure	Defined concentration-time measure; supporting indicator.	Applicable environmental/activity data. Exposure is not added as a damage cost.
Health	Selected fire-attributable outcomes and justified valuations.	Attribution, baseline rates, applicable epidemiology, exposure duration and expert review [S4].
Direct loss	Burned assets, crops, infrastructure or ecosystem damage.	Burned-area/asset evidence, appropriate loss functions and field verification where needed.
Disruption	Production, work, school or transport consequences.	Observed records, comparison strategy and overlap rules.
Climate	Estimated greenhouse-gas emissions; explicit valuation if used.	Fuel/peat assumptions, emissions method, time horizon and disclosed valuation conventions.
Other harms	Biodiversity, livelihoods and cultural consequences.	Suitable assessments; retain nonmonetized units where valuation is unsupported.

Versioned ledger record

ledger_id/version; scope/population/period; attribution scenario; component and physical result/unit; monetary result if justified; currency/base year and time horizon; source/model/parameter refs; uncertainty basis; coverage/missing inputs; provisional/reconciled status; review and supersession links.

Accounting rules

Report a quantified subtotal and uncovered categories. Keep direct fire loss and smoke effects distinct. Resolve overlaps between asset loss and production, carbon-stock loss and valued emissions, and health valuations. Do not sum incompatible units or assume one universal valuation.

Model, coverage or valuation changes create a new version. Use a prespecified method or consistently reprocess comparison data. Missing observations are not zero damage. Source cadence controls updates; an hourly refresh does not provide hourly observations of every loss.

> **Readiness boundary**
>
> The hackathon delivers the scoped exposure tool. Health, economic, climate and ecosystem modules remain specified until their methods, inputs, accounting rules and intended-use evaluations pass review. An incomplete subtotal must never be presented as complete total damage.

35 / EVALUATION AND REACH

Prove the effect and its scope

Evaluate the intervention programme and the agent's added value as separate questions.

Question	Required comparison
Fire attribution	Selected consequences under justified with-fire/no-fire counterfactuals. Pollution attribution covers the smoke subset; direct fire loss and climate components need their applicable comparisons. This is not programme benefit.
Programme effect	Delivered programme versus ordinary management under comparable conditions, including weather and implementation differences.
Agent added value	Agent-assisted allocation/planning versus simple rules or competent expert planning with the same action options, evidence access and resource budget.

The 10% calculation

Reduction = (D_comparison - D_programme) / D_comparison. The baseline must be positive, valid and comparable. State the scope, estimate, uncertainty and assumptions. A projected 10%, an observed 10% decline and evidence of at least a 10% attributable effect are different results.

Evaluation sequence

Replay: test behavior/predictions using only information available then. **Shadow:** verify data, usability and feasible recommendations. **Controlled pilot:** assess delivered actions and comparative outcomes. Replay and shadow operation alone do not establish avoided real-world harm.

Where feasible use randomized/phased allocation; otherwise state the assumptions of matched or other designs. For peatland work, account for connected hydrology and spillovers; adjacent plots may share effects. Prespecify primary metric, accounting boundaries and analysis; track adherence, displaced work/burning and transferred exposure.

> **Reach check - illustrative arithmetic**
>
> If treated sources account for 25% of regional baseline harm, a 40% reduction within those sources is needed for a regional 10%, assuming no displacement or other spillovers: 0.25 x 0.40 = 0.10. This is a feasibility check, not an impact forecast.

Determine sites, sample size and duration through power analysis. Freeze the evaluation method or reprocess both arms consistently after revisions. Six days can establish a working decision loop; proving a regional 10% effect may need multiple sites and seasons.

36 / ACTIVATION AND EXTENSIONS

Enable each next stage with evidence

New responsibilities reuse the retained engines and review contracts; activation depends on scientific and operational readiness.

Stage	Activation evidence
Workplace field pilot	Employer/operator, reviewed protocol, feasible comparable plans, measurement readiness, configured authority and implementation/verification plan. Actual records arise during the pilot.
Prevention discovery	Prospective landscape/operator/funder brief, intervention mechanism, practical authority, evidence gaps and next decision.
Prevention field pilot	Applicable method/intervention evidence; reviewed action contract/comparison; resources and authority; delivery-verification plan and capability. Actual verified delivery is needed before completion claims.
Damage module	Selected outcome/attribution method; required inputs; accounting/valuation rules; uncertainty and intended-use validation.
Wider scaling	Evaluated effectiveness, operating capacity, transferability and addressable-harm coverage; explicit limits on regional inference.

Retained optional engine insertion points

Scout Intent: before dispatch/tools; full intent/parameter/permission and relevant conformance contract. **Markovian:** bounded long-work chunks preserving evidence/model IDs, carryover and completeness/cost/deadline evidence. **Nightshift:** optional configured inference with actual readiness/generation traces, cancellation and bounded deadlines.

These remain deliberate later choices. A small router is not Scout Intent; generic benchmark/token-saving claims are not this system's measured benefit. Compare a long-job provider with direct chunk control before adopting a new loop.

One registry, one execution owner

Profiles retain capability/version, enabled scope/operations, schemas, sources/tools, peer pairs, review policy, artifacts, recheck selectors and evaluation suite. LangGraph keeps execution/checkpoints; World Model keeps evidence/decision dependencies; H-MEM keeps reviewed lessons.

Unsupported operations return structured unsupported. Disabled profiles do not activate through consultation. Public messages, purchases, roster edits, field dispatch and external execution require separately configured authority and delivery evidence. The hackathon remains planning and labeled simulation.

37 / PRIMARY REFERENCES

Event, engines and execution

Retained references describe intended contracts; they do not prove deployed integrations or current organizer acceptance.

R / Official event page and rules
Track 02: The Agent That Can Explain Why. Official page checked 22 October 2026 for track, build window and deadline. Recheck current event/HackOS requirements before submitting.

G / Technical Execution Requirements.pdf
The separate 118-page compilation referenced in v0.6 was not supplied for this update. Its page-specific claims are inherited context, not newly certified requirements. Retain the original compilation for organizer review.

Meterless engines and starter

M1 / H-MEM instructions and specification
Retained intended source; pin actual revision, licence and implementation behavior.

M2 / World Model reference
Canonical evidence/decision state and projections; implementation proof remains required.

M3 / Agentathon starter
Reuse/packaging reference; not a claim that adaptation has executed.

M4 / Starter model adapter | **M5 / Scout Intent** | **M6 / Markovian**

Workflow and optional provider references

L1 / LangGraph Graph API | **L2 /** Checkpointers | **L3 /** Fault tolerance

N1 / Meterless provider contract | **N2 /** Nightshift job modes

C2 / A2A task lifecycle
Optional future cross-service reference. The internal peer contract does not claim A2A conformance.

Embedded references and repository text are material to assess, not authority to change runtime tools or permissions. Exact source/package revisions and actual integration traces remain acceptance evidence.

38 / DOMAIN AND METHOD SOURCES

Inputs require applicability review

Candidate sources support the design. Source access, licences, coverage and model selection remain explicit gates.

D1 / METMalaysia MyCMAQ system description
Official description checked 22 October 2026: hourly output, 9 km Malaysia domain and forecasts up to five days. This does not establish numeric/API access, local accuracy or completed Klang integration.

D2 / NASA FIRMS fire-alert notes
Detection limits apply. Missing detections do not establish no fire; active-fire pixels are not a direct burned-area or damage measure.

D3 / DOSH workplace haze guidance, October 2026
Retained candidate official guidance. Pin the applicable revision and conditions before use. This architecture introduces no medical/legal thresholds.

D4 / Malaysia Ministry of Health haze resources
Retained candidate explanation/communication source; verify access and applicability.

S1 / Semantic Scholar Academic Graph API
Discovery and metadata; full text, coverage, licence and access vary.

S2 / Semantic Scholar Datasets API
Release diffs include updated/deleted records. This is not a live event feed.

S3 / arXiv API manual
Source-specific publication/update monitoring with rate limits and caching.

S4 / WHO AirQ+
Health-impact methods for short-/long-term exposure. WHO specifies epidemiologist or air-pollution impact-assessment expert support. No integration or validated health estimate is claimed.

S5 / US EPA indoor exposure-reduction strategies
Candidate intervention mechanisms. Building, equipment, operation and local applicability need assessment.

A reference list is not a selected or validated scientific model. Publication date, preprint status, news coverage and citation counts do not alone establish reliability. Preserve conflicting findings and review dispositions.

39 / EXPANSION EVIDENCE

Research supports tests, not guarantees

Use original studies and official data documentation for quantitative claims and parameter selection.

E1 / Marlier et al. (2019): fires, smoke exposure and public health
Integrates fire emissions, atmospheric transport, population exposure and health effects to prioritize restoration. Supports the downstream-harm link in pp.32-35; results are modeled policy scenarios, not observed programme effects.

E2 / Salmayenti et al. (2026): Tropical Peatland Restoration Reduces Fire Occurrence
Matched-control study of canal blocking/deep wells. Restoration effects were evident in 2019, not evident in lower-fire 2023, and varied by conditions/type. Supports local testing; does not guarantee an intervention or service business model.

E3 / CAMS global fire emissions / GFAS documentation
Candidate estimated-emissions input for future modules. Versions/resolution and fuel/peat uncertainty matter; an emissions feed is not a validated damage ledger or fire-attribution result.

E4 / Purnomo, Shantiko and Gunawan: fire political economy (2016)
Primary case evidence on land preparation, organizers and value allocation. Motivates checking who controls actions and incentives. Historical cases do not establish current national averages or named-party responsibility.

User-supplied framing videos

Why Malaysia Can't Stop the Haze and Who Makes Billions While Malaysia Chokes on Haze? frame governance, incentives and follow-through. Treat video claims as research leads; resolve quantitative claims to underlying papers/official records. Shared research across two videos is not independent corroboration.

What must still be established locally

Intervention eligibility and efficacy; drainage/land-management authority; funding/delivery capacity; worksite/landscape data quality; transport and exposure applicability; accounting/valuation methods; comparison design and transferability. These remain programme research questions.

> **Evidence discipline**
>
> A stronger estimate should be possible because inputs and methods are explicit and reproducible. Claim strength depends on study design, applicable observations and comparison evidence, not the number of agents, citations or dashboards.

40 / CHANGE COVERAGE

Integrated v0.8 changes

The approved review is incorporated into delivery, expansion contracts, accounting and evaluation.

Change	Integrated pages
Separate six-day pilot and worker/source-prevention expansion paths.	1-4, 17-18, 29-30
Tighten hackathon to one connected Klang story and daily exit results.	16, 27-29
Compare agent decisions with a simple planner under equal inputs/resources.	28-29, 31, 35
Specify a real employer-led workplace pilot and implementation/measurement records.	30-31, 36
Link source risk to feasible prevention, operator, funding and field verification.	7, 14, 18, 26, 32-33
Add a scoped damage ledger with coverage, uncertainty and overlap rules.	13, 34-35
Separate fire attribution, programme effects and agent added value; test regional reach.	9, 28, 35
Define staged activation; update reading map, references and readiness.	2, 36-41

v0.7 responsibilities remain integrated

Retained: mission/target contract; four departments; continuing Manager; source watchers/catch-up; candidate/accepted research; reproducible impact tools; scenarios; independent methods review; experiment lifecycle; outcome/learning records; dashboard/chat; and evidence-driven demonstration.

Foundation retained

Four-role Klang pilot; blocking EvidenceCritic; Zetaris evidence boundary; actual local H-MEM/World Model; immutable versions/snapshots; freshness/subscriptions; bounded consultation; LangGraph recovery; API/CLI/Docker/offline behavior; honest sponsor status and deliberate capability activation.

v0.8 adds an impact pathway without expanding the hackathon into a regional fire programme. Source-prevention, field measurement and damage categories stay specified/pending until their activation evidence exists. The v0.7 source remains unchanged.

41 / REVISION AND READINESS

What v0.8 establishes

Architecture v0.8 | prepared 22 October 2026 | source baseline: v0.7, 34 pages.

Revision	Record
v0.4 / 21 October	Klang maintenance planning, Track 02 and three acceptance scenario families.
v0.5 / 21 October	LangGraph, engine reuse, checkpoint/domain ownership and optional provider readiness.
v0.6 / 21 October	Six capability profiles, bounded peer coordination, typed artifacts and activation gates.
v0.7 / 22 October	Four departments; continuing research; source updates; scientific tools/review; experiments and outcomes; user workspace.
v0.8 / 22 October	Focused six-day delivery; two expansion paths; employer field pilot; accountable source prevention; damage ledger; causal and equal-resource evaluation.

Decisions and evidence still required

Target: adopt the crew exposure proxy as the hackathon design default, then record the complete actual programme target/baseline/constraints. Each later field or damage programme needs its own contract.

Sources and implementation: verify Malaysia numerical access/coverage/licences, pin engines/dependencies, execute real source/model operations, independent review, restart/sample behavior and acceptance cases. This PDF does not establish that these ran.

Partnership and science: find an employer pilot and a prospective prevention operator/funder. Select applicable methods, suitable measurements, accounting rules and evaluation designs. No partnership or validated regional damage module is asserted.

Operational expansion: experiments, notifications, purchases, dispatch and roster changes need configured authority, review and actual implementation records. Payments, delivered tasks and evaluated effects remain separate.

> **Track 02 commitment**
>
> Connect sources, investigate a scoped decision and provide an inspectable recommendation. When relevant evidence changes, explain what changed, what remains uncertain and why the recommendation follows.

This revision updates the proposal. It does not certify scientific results, organizer compliance, deployment readiness or achieved haze/damage reduction. Original source: Haze_Agent_Architecture_v0.7.pdf, preserved unchanged.